

Measuring Performance in Capital-Light Businesses

In this book, our primary measure of return on capital is return on invested capital (ROIC). We define ROIC as net operating profit after taxes (NOPAT) divided by invested capital. We derive ROIC from items on a company's financial statements, with some adjustments, such as separating operations from financing and separating operating items from nonoperating items.¹ ROIC correctly reflects return on capital in most cases, but special circumstances require alternative measures. For example, a young biotech company could spend a billion dollars on research and development (R&D) before its product is launched. Since R&D is expensed, not capitalized, the company would show a negative ROIC in its early years and a very high ROIC once the product is launched. The actual economic return on capital over the life of the product would lie at some average level in between.

In this chapter, we show how to deal with such investments in R&D and in marketing and sales that are expensed when they are incurred. Creating pro forma financial statements that capitalize these expenses can provide more insight into the underlying economics of a business. In addition, we discuss businesses with very low capital requirements, where we recommend using economic profit or economic profit scaled by revenues to measure return on capital.

CAPITALIZING EXPENSED INVESTMENTS

When a company builds a plant or purchases equipment, it capitalizes the asset on the balance sheet and depreciates it over time. Conversely, when a company invests in intangible assets such as a new production technology, a brand name, or

¹ In Chapter 11, we explain why we use ROIC instead of other accounting-based metrics like return on equity (ROE) or return on assets (ROA).

a distribution network, the entire outlay must be expensed immediately. In sectors such as pharmaceuticals, high technology, and branded consumer goods, failure to recognize such expenses as investments can lead to significantly underestimating a company's invested capital and overstating its return on invested capital.

To get a more accurate measurement of ROIC,² it's best to capitalize outlays for intangible investments if they bring benefits over multiple years in the future rather than merely for the current year. Earnings in any given year are supported by not just that year's R&D or brand advertising expenses, but instead by many prior years of these expenses. It has taken companies such as Coca-Cola and PepsiCo many decades and billions of dollars to build their global brand names. Pharmaceutical companies such as Pfizer, and high-tech companies such as Intel and ASML, had to invest in technology development projects over many years to build and sustain their current product offerings.

The economics of investments in intangible assets are very similar to those of investments in tangible assets. Their treatment in ROIC should therefore also be the same to ensure that it adequately reflects the internal rate of return (IRR), or true return, of the underlying investments.³ Failure to do so would lead to ROICs far above the true return of the business. Consider what would happen to ROIC if capital expenditures for net property, plant, and equipment (net PP&E) were not capitalized but were expensed instead.

In addition to improving the measurement of ROIC, capitalizing intangible investments can reduce the manipulation of short-term profits. Under traditional accounting, a manager looking to meet short-term earnings targets can simply reduce R&D spending. With R&D capitalized, however, amortization charges to earnings will remain almost unchanged in the short term. Capitalizing investments can also provide strategic insights. For example, many companies set R&D budgets at a fixed percentage of revenue. When combined with expensing R&D, this masks the change in performance resulting from any change in revenues, because the earnings margin remains unchanged. But when R&D is capitalized, amortization charges do not change with revenues, and the impact on performance is clearly reflected in earnings.

Example: Capitalizing R&D Expenses

As an illustration of capitalizing intangible investments and its impact on ROIC, Exhibit 24.1 presents the reorganized financial statements for PharmaCo. This fictional company has experienced rapid growth over the past 25 years, reaching around \$1.2 billion in revenues by 2020. The after-tax earnings margin is 11 percent of sales. R&D expenses, to renew the product pipeline, are at around 20 percent of sales. ROIC is at 33 percent, with revenues at three times invested capital as

² The same applies to return on capital measures such as cash flow return on investment (CFROI), as discussed in the following chapter.

³ To be truly "value based," the measure for return on capital should reflect the internal rate of return (IRR) of the underlying business from the time investments are made until all the cash flows from that investment have been collected (see also Chapter 25).

EXHIBIT 24.1 **PharmaCo: Reorganized Financial Statements**

\$ million

Partial income statement	2015	2016	2017	2018	2019	2020	
Revenues	1,045	1,077	1,109	1,142	1,176	1,212	
Cost of sales	(627)	(646)	(665)	(685)	(706)	(727)	← Fixed at 60% of revenues
R&D expense	(229)	(235)	(242)	(248)	(255)	(262)	
Operating profit	189	195	202	208	215	222	
Taxes	(76)	(78)	(81)	(83)	(86)	(89)	
NOPAT ¹	113	117	121	125	129	133	

Partial balance sheet	2015	2016	2017	2018	2019	2020	
Invested capital	348	359	370	381	392	404	← Fixed at 3 times capital turnover
NOPAT/revenues, %	10.9	10.9	10.9	10.9	11.0	11.0	
ROIC, %	32.6	32.7	32.8	32.8	32.9	33.0	

¹ Net operating profit after taxes.

computed directly from the balance sheet. But this ROIC does not represent the company's true economic performance, because the invested capital includes only purchased capital and not the intellectual capital created internally from R&D.

To estimate ROIC with capitalized investments in R&D, use the following three-step process:

1. Capitalize and amortize the R&D asset, using an appropriate asset lifetime.
2. Adjust invested capital upward by the historical cost of the R&D asset, net of cumulative amortization.
3. Adjust NOPAT by replacing R&D expense with R&D amortization. (Do not adjust operating taxes.)

To capitalize the R&D asset, choose a starting year, and begin accumulating R&D expenses. Choose the earliest year feasible, as the model requires accumulated R&D to reach a steady state before the adjusted ROIC calculation becomes meaningful. Exhibit 24.2 starts in 1995, assuming straight-line amortization and an eight-year R&D asset life. PharmaCo spent \$22 million on R&D in 1995, which we capitalize and add to invested capital and start to amortize in 1996. By adding R&D expenses to the prior year's net asset value and then deducting amortization charges in each year, we arrive at a capitalized R&D asset base of \$1,666 million in 2020.⁴

To adjust invested capital for the intangible investments, add the capitalized R&D asset to invested capital. On this basis, PharmaCo's total capital amounts to \$2,070 million in 2020, most of it in the form of capitalized R&D.⁵

⁴ In this example, for illustration purposes, we approximate amortization at 10 percent of the preceding year's ending balance. Advanced models use straight-line amortization of actual R&D expense.

⁵ If we add capitalized R&D to operating assets, total funds invested will no longer balance. To balance total funds invested, add capitalized R&D to equity equivalents. For more on total funds invested and their reconciliation, see Chapter 11.

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EXHIBIT 24.2 PharmaCo: Capitalization of R&D

\$ million

Estimated R&D asset lifetime: 8 years							
Partial income statement	1995	1996	1997		2018	2019	2020
Revenues	10	22	43	...	1,142	1,176	1,212
R&D expense	(22)	(24)	(29)	...	(248)	(255)	(262)
Capitalized R&D asset							
Capitalized R&D, starting	—	22	44	...	1,477	1,541	1,604
R&D expense	22	24	29	...	248	255	262
Amortization	-	(3)	(5)	...	(185)	(193)	(200)
Capitalized R&D, ending	22	44	67	...	1,541	1,604	1,666
Partial balance sheet							
Invested capital, unadjusted	3	7	14	...	381	392	404
Capitalized R&D	22	44	67	...	1,541	1,604	1,666
Invested capital, adjusted	25	51	81	...	1,922	1,996	2,070

Adjust NOPAT by replacing R&D expense (\$262 million in 2020) with R&D amortization (\$200 million), computed as outlined in Exhibit 24.3. Operating taxes remain unchanged, because capitalization and amortization of R&D expense does not change taxable income for fiscal purposes. For PharmaCo, replacing R&D expense with amortization raises NOPAT in 2020 from \$133 million to \$195 million. This is quite common for growth firms, as current R&D is typically higher than the amortization of historical R&D. As the company's growth rate tapers off, however, amortization will catch up with expense, and NOPAT adjustments will be small.

EXHIBIT 24.3 PharmaCo: NOPAT Adjusted for R&D Capitalization

\$ million

	2016	2017	2018	2019	2020
Revenues	1,077	1,109	1,142	1,176	1,212
Cost of sales	(646)	(665)	(685)	(706)	(727)
R&D expense	(235)	(242)	(248)	(255)	(262)
Operating profit	195	202	208	215	222
Operating taxes	(78)	(81)	(83)	(86)	(89)
NOPAT	117	121	125	129	133
Add back: R&D expense	235	242	248	255	262
R&D amortization	(168)	(177)	(185)	(193)	(200)
Adjusted NOPAT	184	186	189	192	195
ROIC, %	32.7	32.8	32.8	32.9	33.0
ROIC adjusted for R&D capitalization, %	10.4	10.1	9.8	9.6	9.4

EXHIBIT 24.4 **PharmaCo: Free Cash Flow**

\$ million

	R&D expensed, unadjusted	2017	2018	2019	2020
NOPAT		121	125	129	133
Depreciation		37	38	39	40
Gross cash flow		158	163	168	174
Capital expenditures		(48)	(49)	(51)	(52)
Free cash flow		110	114	118	122

	R&D capitalized	2017	2018	2019	2020
Adjusted NOPAT		186	189	192	195
Depreciation		37	38	39	40
→ Amortization of R&D		177	185	193	200
Gross cash flow		400	412	424	436
Capital expenditures		(48)	(49)	(51)	(52)
→ Investment in R&D		(242)	(248)	(255)	(262)
Free cash flow		110	114	118	122

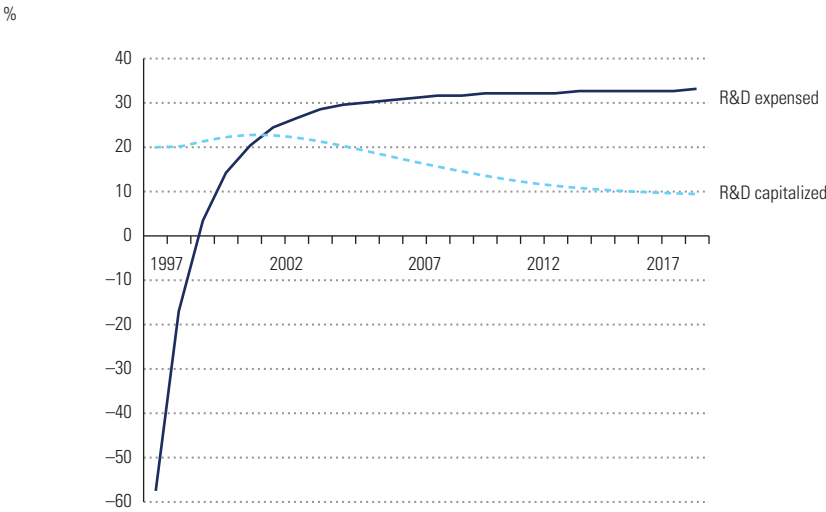
Note that for PharmaCo's historical years, free cash flows cannot change when R&D expenses are capitalized (see Exhibit 24.4). The amortization is a noncash charge in NOPAT and is added back to calculate gross cash flow. This effectively moves R&D expenses from gross cash flow to investments, leaving free cash flow unchanged.

Based on the new measures for invested capital, with capitalized R&D investments and for NOPAT with R&D amortization instead of expenses, we derive an adjusted ROIC. The adjusted ROIC with R&D capitalized represents PharmaCo's return on capital, including intangible investments. It can be compared with an unadjusted ROIC with R&D expensed, as shown in Exhibit 24.5. Because the R&D asset lifetime was estimated at eight years, at least as many years of constant growth must elapse for capital and ROIC to reach a steady state and provide a meaningful indication of true economic returns. As Exhibit 24.5 shows, the adjusted ROIC computed on total capital stabilizes at around 9.5 percent, dramatically lower than the 33 percent ROIC derived from the unadjusted financial statements. As long as the R&D investments needed to support earnings remain unchanged, PharmaCo's adjusted ROIC is the better estimate of its true economic return and underlying performance.⁶

One of the key assumptions made in capitalizing intangible investments is the asset lifetime. Although it may be hard to come up with an accurate estimate, this should not keep you from capitalizing the R&D expenses. Asset lifetime has less impact on ROIC than you might expect. In the PharmaCo example, we

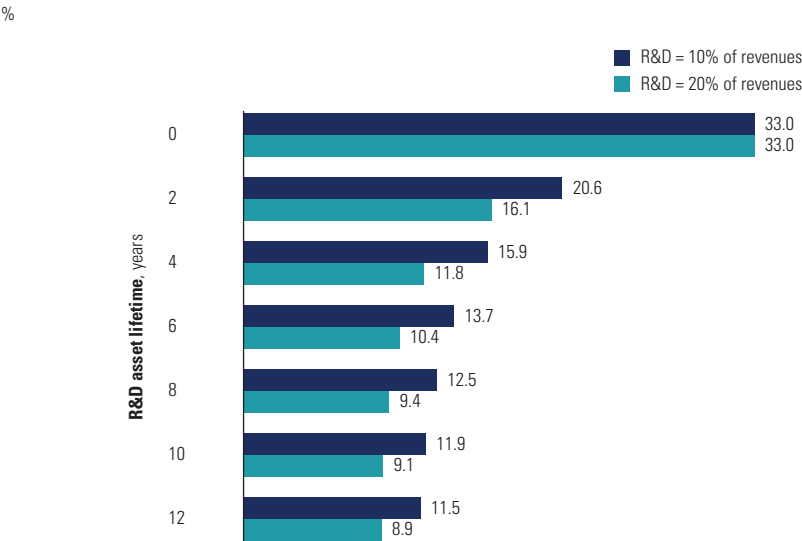
⁶ That is, ROIC is the better estimate of the investments' value creation, as explained in Chapter 25.

EXHIBIT 24.5 **PharmaCo: ROIC, 1997–2020**



assumed an asset life of eight years. In Exhibit 24.6, we stress-test this assumption by varying asset life between two and 12 years. Even an asset life of just two years dramatically reduces PharmaCo’s ROIC from 33 percent when R&D is expensed to 16 percent when it is capitalized. Increasing the asset life continues to lower ROIC, but by smaller amounts as asset life increases. So choosing an asset life of 12 rather than eight years (a reasonable range for the life of most R&D

EXHIBIT 24.6 **PharmaCo: ROIC at Different Estimates of R&D Asset Lifetime, 2020**



assets) does not materially affect perceptions of performance: for PharmaCo, ROIC would be 8.9 percent for a 12-year life, versus 9.4 percent for an eight-year life. This pattern remains unchanged when R&D spending is much lower—for example, at only 10 percent of revenues. Furthermore, when using ROIC to compare the performance of competing companies, what matters most is that asset lifetime estimates are consistent across all companies. Keep in mind that the lifetimes for tangible assets are also based on rough estimates and accounting conventions. Yet most managers and analysts are quite comfortable using tangible-asset book values and depreciation charges as the basis for return on capital and earnings.⁷

Interpreting Return on Capital, Including Capitalized Expenses

In general, capitalizing intangible investments will lead to lower ROIC. For mature companies with stable revenues and investment spending, the amortization charges for intangible assets are likely to be close to the amounts expensed. As a result, capitalizing the expenses may have little impact on NOPAT. But the capital base will always increase when the expenses are capitalized, leading to lower ROIC.

Although the capitalization can never change historical free cash flows, as discussed in the PharmaCo example, the resulting adjustments to capital turnover and ROIC can affect projections of future free cash flows. For PharmaCo, required investments in R&D to achieve growth of 10 percent per year would be estimated at \$375 million in 2021, which is \$113 million more than the \$262 million spent in 2020. This follows from required growth of the net R&D asset base (10 percent, or \$167 million) plus an annual amortization charge of \$208 million (one-eighth of the 2020 ending balance). When R&D investments going forward would be modeled as expenses, the required additional R&D outlay in 2021 would be only 10 percent of the additional 2021 revenues, or \$26 million. This is comparable to what happens to investment projections if capital expenditures for tangible assets are derived from a constant ratio to revenues or instead implied from a constant capital turnover (see Chapter 13).

If PharmaCo can increase its revenues by 10 percent as a result of increasing its R&D expenses by 10 percent, the unadjusted ROIC provides the best estimate of the IRR of future investments in its business. In contrast, if achieving that same revenue growth would require PharmaCo to increase its net R&D asset base, rather than its R&D expenses, by 10 percent, the adjusted ROIC is the better estimate. Of course, these R&D investment estimates for PharmaCo are not likely to apply from year to year. What matters is which R&D investments are required for growth over the long term.

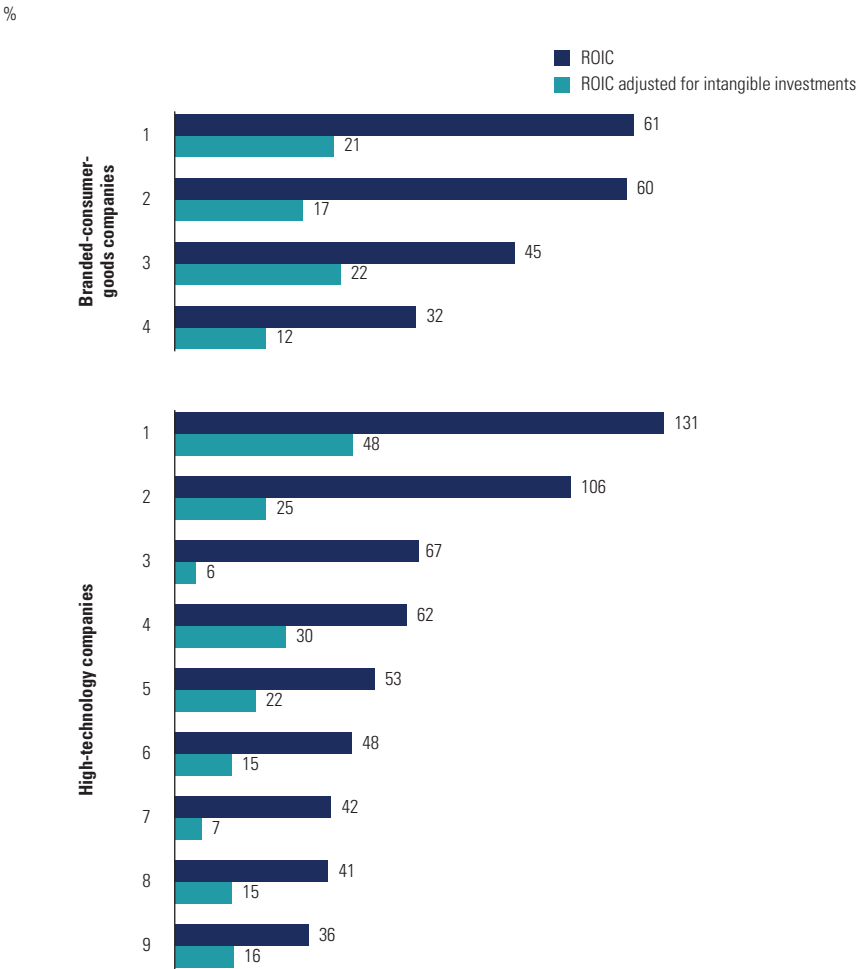
More accurately reflecting the economics of intangible investments on ROIC can have major implications for investment decisions, performance

⁷ Note also that for an alternative measure of ROIC, such as cash flow return on investment (CFROI) with or without resource capitalization, estimates of asset lifetimes are critical—not for book value or depreciation, but for estimating the CFROI itself (see Chapter 25).

assessments, resource allocation, and competitive behavior. For instance, if the cost of capital is 10 percent, PharmaCo is in fact destroying value, and management should question continued investment. Competitors should question the validity of entering the company's product markets. The margins may be high, but required investments in R&D are large.

To illustrate the impact of capitalizing intangibles on estimates of ROIC and perspectives on value creation, we analyzed past spending on research and advertising over a ten-year period for four global companies in branded consumer goods. After the estimated past expenses in R&D and advertising were capitalized and amortized, ROIC for all companies decreased significantly and also provided a very different ranking of performance, as shown in the top portion of Exhibit 24.7. A similar analysis of ROIC including capitalized

EXHIBIT 24.7 **Impact of Adjusting ROIC for Intangible Investments**



R&D expenses among high-tech hardware manufacturers provided similar shifts in perceived performance levels and rankings (see the bottom portion of Exhibit 24.7).

Capitalizing intangibles can provide a better financial perspective on competitive positions. Think of comparing current budgets on brand advertising between incumbents and new entrants in personal or household products. The comparison is not very useful if the incumbent brands have been built by many years of marketing efforts. Incumbents' current advertising budgets will then underestimate the investments required by new entrants to reach similar levels of brand awareness among customers. A capitalized investment base can provide a more accurate estimate.

While insights from capitalizing resources are valuable, companies must take care. Left unchecked, managers could have an incentive to classify all expenses as investments, even those with no long-term benefits, because this will maximize reported short-term performance. They could also be reluctant to write off investments that prove worthless after they have been capitalized. For instance, a distribution channel may be kept open merely to avoid a write-down on the manager's economic balance sheet.

WHEN BUSINESSES NEED LITTLE OR NO CAPITAL

Some businesses do not require significant amounts of capital—for example, those in the professional services sector, but also consumer electronics companies with outsourced manufacturing. Because of these companies' low or even negative capital base, ROIC can become less meaningful. In such cases, we recommend using economic profit as the key measure of value creation.

Capital-Light Business Models and ROIC

Examples of businesses with an inherently low need for capital include accounting, legal counseling and other professional services, and real estate and other forms of brokerage services. Businesses such as software development and services have limited fixed capital needs, and customer license prepayments and supplier financing often bring their overall invested capital close to zero. In these cases, capital is very low relative to earnings generated, and ROIC accordingly is high. Modest changes in an already small invested-capital base can lead to very large swings in ROIC, making ROIC in any particular year hard to use for performance management or financial planning and target setting.

Let's illustrate with a stylized example of TradeCo, whose financial statements are summarized in Exhibit 24.8. TradeCo is a trading company in plumbing supplies and tools. It has offices and a warehouse in a low-cost location. Inventories are kept to a minimum: except for those items with the

EXHIBIT 24.8 TradeCo: Financial Statements

\$ million

	NOPAT	2015	2016	2017	2018	2019	2020
	Revenues	200.0	209.0	212.1	216.4	214.2	212.1
	Cost of goods sold	(160.0)	(165.1)	(169.7)	(175.3)	(171.4)	(170.7)
	SG&A	(20.0)	(20.9)	(21.2)	(21.6)	(21.4)	(21.2)
	Operating taxes	(7.0)	(8.0)	(7.4)	(6.8)	(7.5)	(7.1)
	NOPAT	13.0	14.9	13.8	12.7	13.9	13.1
	Invested capital	2015	2016	2017	2018	2019	2020
	Net working capital	(12.0)	(8.4)	(10.6)	(2.2)	(4.3)	(6.4)
	Net PP&E	10.0	9.5	9.1	9.0	8.7	8.4
	Invested capital	(2.0)	1.1	(1.5)	6.8	4.4	2.1
	Free cash flow	2015	2016	2017	2018	2019	2020
	NOPAT	13.0	14.9	13.8	12.7	13.9	13.1
	Net investments	(2.0)	(3.1)	2.6	(8.3)	2.4	2.3
	Free cash flow	11.0	11.9	16.3	4.4	16.3	15.4
	Key value drivers, %	2015	2016	2017	2018	2019	2020
	NOPAT/revenues	6.5	7.2	6.5	5.9	6.5	6.2
	Invested capital/revenues	(1.0)	0.5	(0.7)	3.1	2.1	1.0
→	ROIC	N/M ¹	1,371	N/M	186	316	632
	Economic profit	2015	2016	2017	2018	2019	2020
	NOPAT	13.0	14.9	13.8	12.7	13.9	13.1
	Capital charge ²	0.2	(0.1)	0.1	(0.7)	(0.4)	(0.2)
→	Economic profit	13.2	14.8	13.9	12.0	13.5	12.9

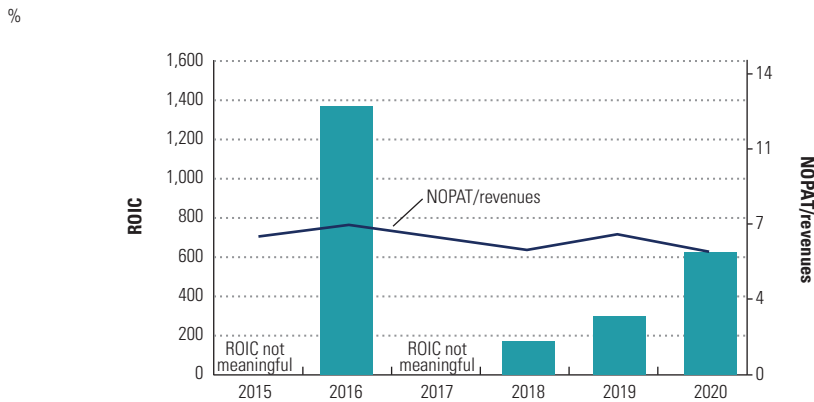
¹ Not meaningful.² Cost of capital equals 10%.

highest turnover, supplies and tools are purchased on customer order. Because TradeCo pays its suppliers after receiving payment on its own customer invoices, working capital is negative.

As Exhibit 24.8 shows, revenues, earnings, and free cash flow are fairly stable on a year-by-year basis. But as the graph in Exhibit 24.9 shows, ROIC fluctuates wildly and is even unmeasurable in some years, despite stable earnings margins and healthy cash flows. The reason is that TradeCo's invested capital is very small and sometimes even negative, mainly because of movements in working capital. ROIC is not meaningful in 2015 and 2017 because the company had negative invested capital. ROIC is numerically negative, but it lacks any economic interpretation.⁸ Looking at the bottom of Exhibit 24.8, we see that economic profit was positive in 2017, clearly indicating value creation. The movements in ROIC could mislead your assessment of performance. For example, ROIC increased from 316 percent in 2019 to 632 percent

⁸ Mathematically, ROIC still ties perfectly with cash flow and value, following the fundamental logic described in Chapter 3.

EXHIBIT 24.9 TradeCo: ROIC and NOPAT Margin



in 2020. Yet value creation declined, as the change in economic profit for the same period shows. The change in ROIC was driven by a decline in working capital. Earnings declined simultaneously and pushed down value creation.

Not all businesses with low capital are inherently capital light. Indeed, some capital-intensive businesses have adopted capital-light models by outsourcing their most capital-intensive processes—typically manufacturing and distribution. The high-tech electronics sector provides examples of this approach, including Apple, Fujitsu, Hewlett-Packard, and Sony. In the apparel sector, companies such as Nike have outsourced their manufacturing.

ROICs for businesses that have aggressively outsourced parts of their business chain can be very high and volatile. In addition, the capital reduction that comes with outsourcing can lead to confusion when ROIC is used to assess whether outsourcing creates any value to begin with. After outsourcing, many businesses end up with much higher ROICs. In some cases, managers even refer to the higher ROIC as one of the main benefits of outsourcing. But the ROIC increase does not necessarily mean that the company has created value for its shareholders.

Consider the companies InhouseCo and ContractCo in Exhibit 24.10. The companies are identical, with one exception: ContractCo has outsourced all of its production to a third party. It has no net PP&E and no depreciation charges, but it has higher operating costs compared with InhouseCo. Although ContractCo's earnings are lower than InhouseCo's, its ROIC is more than five times larger because it no longer needs PP&E. But ContractCo is not creating more value in its business than InhouseCo. In fact, as the measure of economic profit indicates, the two companies' value creation is identical. In this example, ContractCo has separated out its capital-intensive and low-ROIC production activities from its other activities without creating value. The ROIC for ContractCo goes up simply because it retains only the high-ROIC activities. But that does not say anything about the value creation from

EXHIBIT 24.10 Impact of Production Outsourcing on ROIC

\$ million

	InhouseCo	ContractCo
NOPAT		
Revenues	100.0	100.0
Operating costs	(85.0)	(94.5)
Depreciation	(3.8)	—
Operating taxes	(3.9)	(1.9)
NOPAT	7.3	3.6
Invested capital		
Net working capital	5.0	5.0
Net PP&E	50.0	—
Invested capital	55.0	5.0
Key value drivers, %		
NOPAT/revenues	7.3	3.6
Invested capital/revenues	55.0	5.0
→ ROIC	13.3	71.2
Economic profit		
NOPAT	7.3	3.6
Capital charge ¹	(4.1)	(0.4)
→ Economic profit	3.2	3.2

¹ Cost of capital equals 7.5%.

outsourcing.⁹ Managers should therefore not make decisions to outsource merely on the grounds that it raises ROIC. These decisions need to be supported by an analysis of economic profit or, equivalently, a DCF valuation.

Economic Profit as a Key Value Metric

Although there is no objective way to determine a cutoff point, we believe that ROICs above 50 percent need to be handled with caution when used as a measure of value creation. Special caution is required in businesses where high capital turnover, rather than high earnings margins, drives such ROIC levels.

In such cases, economic profit is a more solid performance measure that is always in line with value creation. (For more details on economic profit, see Chapter 3.) It can be defined in either of the following two equivalent ways:

$$\text{Economic Profit} = (\text{ROIC} - \text{WACC}) \times \text{Invested Capital} \quad (24.1)$$

$$\text{Economic Profit} = \text{NOPAT} - \text{Capital Charge} \quad (24.2)$$

In Equation 24.2, the capital charge equals WACC times invested capital.

⁹ Of course, outsourcing in this example could still create real value if it enables ContractCo to realize higher growth because it needs less capital for its business.

EXHIBIT 24.11 **DiversiCo: Economic Profit Scaled by Revenues**

	Revenues	Invested capital	NOPAT	NOPAT/ revenues, %	ROIC, %	Economic profit ¹	Economic profit/ revenues, ¹ %
Software	100	(5)	25	25	n/m ²	25	25
Hardware services	250	10	44	18	438	43	17
Supplies	750	250	94	13	38	73	10
Hardware	2,500	1,000	188	8	19	103	4

¹ Cost of capital equals 8.5%.² Not meaningful.

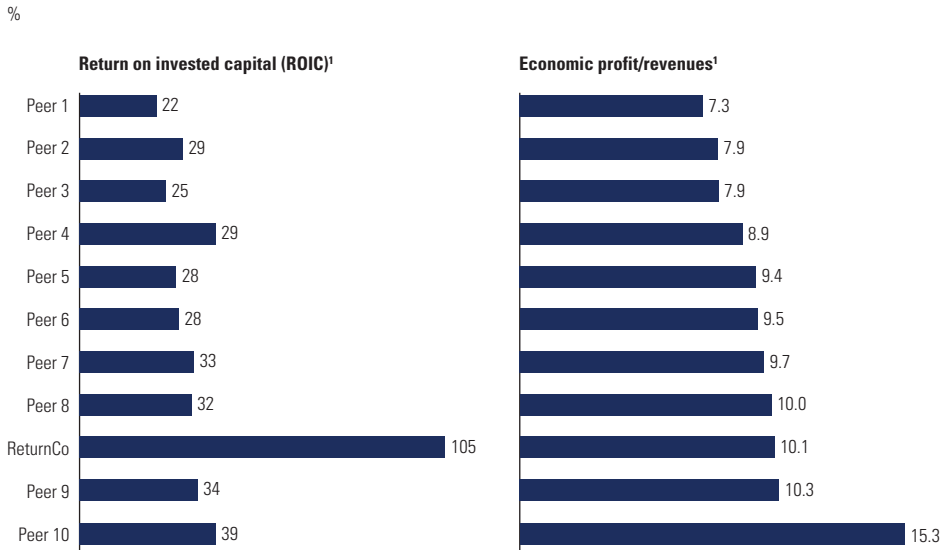
Because ROIC is multiplied by invested capital, economic profit automatically corrects for any distortion in ROIC for business models with extremely low capital intensity. The TradeCo example in Exhibit 24.8 illustrated this. ROIC shows very large fluctuations over the years, even becoming unmeasurable in some years. In contrast, economic profit is fairly stable, just as TradeCo's cash flows are stable and consistently positive over the years. Economic profit is a much better reflection of TradeCo's underlying business economics. It provides more accurate insights into its historical performance and a useful basis for predicting its future performance.

As economic profit is a measure of return on capital in absolute terms, it is very useful for understanding whether value creation in a particular business has increased from one year to the next. But it is harder to use for interpreting differences in economic profit generated by businesses of different sizes. Take, for example, DiversiCo in Exhibit 24.11. DiversiCo is a diversified industrial company with business units in software, hardware, hardware services, and supplies. The business units are very different in size and economics. Hardware, for example, has annual revenues of \$2.5 billion, dwarfing the \$100 million in revenues generated by software development. The software business has negative invested capital, thanks to customer prepayments, whereas hardware requires \$1 billion in capital, mainly for manufacturing and distribution facilities and inventories. ROIC is meaningless for comparing performance across DiversiCo's businesses, because software and hardware services have little or negative capital. Economic profit provides an accurate picture of value creation, but comparisons among businesses of such different sizes are difficult. Economic profit is lowest for the software business (at \$25 million), not so much because of the business's performance, but because of its size.

To better compare the value creation of DiversiCo's businesses, scale economic profit by revenues, turning it into a measure of value creation per dollar of sales.¹⁰ As graphed in the final column of Exhibit 24.11, it now becomes clear that DiversiCo's software business generates the highest value per dollar

¹⁰ See M. Dodd and W. Rehm, "Comparing Performance When Invested Capital Is Low," *McKinsey on Finance* (Autumn 2005): 17–20.

EXHIBIT 24.12 **Better Performance Comparison with Economic Profit over Revenues**



¹ Excluding goodwill and acquired intangibles.

Source: Capital IQ, Corporate Performance Analytics by McKinsey.

of revenues, and its hardware business the lowest. Driving revenue growth in software development would therefore be most beneficial for shareholders.¹¹ Scaling economic profit in this way provides DiversiCo's management with a better yardstick for decisions on resource allocation and portfolio strategy.

In the same way, the ratio of economic profit over revenues can help in benchmarking performance with peers of different size and capital intensity. Consider the example of a branded-consumer-goods company, which we refer to as ReturnCo, in Exhibit 24.12. ReturnCo is generating a ROIC of 105 percent, far above its international peers' ROIC levels of around 30 to 40 percent. But this does not necessarily mean that ReturnCo creates more value and that it has some source of competitive advantage over its peers. Following our rule of thumb, ROICs above 50 percent should be interpreted with caution and carefully analyzed. In this case, it turns out that ReturnCo provides its customers with aggressive discounts for early payment. The discount pushes its earnings margins below peer levels, but the early payments make its net working capital negative and reduce its invested capital. The net result is an exceptionally high ROIC. Comparing its ROIC with those of its peers is pointless because of this difference in capital intensity, and absolute economic profit will of course differ with the size of the competitors. Instead, an analysis of

¹¹ Note how economic profit over revenues is almost identical to NOPAT margin for capital-light businesses, such as software and hardware services in this example. This is easily explained by examining Equation 24.2: when invested capital is 0, the capital charge is 0, and economic profit is equal to NOPAT.

economic profit over revenues best reveals how ReturnCo performs relative to its peers in terms of value creation. As the exhibit shows, the increased capital efficiency is roughly offset by the discount provided: ReturnCo's ratio of economic profit to revenues is very similar to those of its peers. At first sight, ReturnCo's ROIC appeared superior, but a closer look has revealed that its value creation is in line with that of its peers.

In general, when you are comparing the performance of businesses with very different capital intensity and size, using economic profit over revenues provides the best insights into performance and value creation.

SUMMARY

For most businesses, ROIC is a good measure of return on capital. However, for businesses that rely on significant investments in intangibles, such as R&D or brands, you should make some adjustments to ROIC to include the capitalized value of these resources. For businesses that use very little or no capital, economic profit is a better measure of value creation. To allow for comparison across businesses of different sizes, you can scale economic profit by revenues.

REVIEW QUESTIONS

1. Why should managers capitalize expensed items like research and development (R&D) and marketing when estimating their firm's return on invested capital (ROIC)?
2. ResearchCo is a medical devices company, producing equipment for diagnosing and treating heart disease. The company currently generates \$100 million in revenues and is expected to grow 10 percent per year. ResearchCo maintains cost of sales at 50 percent of revenue, maintains R&D at 15 percent of revenue, and pays an operating tax rate of 30 percent. To generate 10 percent growth, ResearchCo will reinvest 20 percent of operating profit each year. Last year's invested capital equaled \$75 million. Using the preceding data, forecast five years of net operating profit after tax (NOPAT) and five years of invested capital. What is the ROIC on year-end capital by year? Do not capitalize R&D.
3. Your colleague argues that R&D for ResearchCo should be capitalized and amortized. If R&D is capitalized, what will be the starting R&D asset, investment in R&D, amortization of R&D, and ending R&D asset by year in the five-year forecast? Use straight-line amortization over three years, with the first amortization charge deducted the following year.
4. Use the R&D capitalization table developed in Question 3 to modify NOPAT and invested capital from Question 2. What is the ROIC on year-end capital by year? How does this compare with the ROIC computed in Question 2?
5. Compute the annual free cash flow for ResearchCo with and without the capitalization of R&D. How do the two sets of free cash flows differ? Assume no depreciation of physical assets.